Q4 NO CALCULATOR ALLOWED Q4

Answer QUESTION 4 PARTS A and B on this page.

PART A

$g'(x) = f(x)$

$g'(8) = f(8)$

$g'(8) = 1$

$f(8)$ appears to be 1 when looking at the graph

PART B

when $x=0$ there appears to be a point of inflection as f goes from concave up to concave down.

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

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Q4 NO CALCULATOR ALLOWED Q4

Answer QUESTION 4 PARTS C and D on this page.

PART C

$g(12) = 9$

$g(0) = -\frac{9}{2}\pi$

PART D

$g'(x) = f(x) = 0$

$x = 0, 6$

minimum when $x = 0$ knowing that $g'(x) = f(x)$ using the FTC, 0 is a minimum as $f(0) = 0$ meaning the slope of g 0 and f goes from negative to positive at that point making it a minimum.

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.